Krishna Defence System / Garuda AI

World-Model Predictive Cyber Defence

Learn the evolving network state, simulate multiple future states, explain the risk trajectory, and route evidence into controlled defensive action.

Krishna Defence System is the complete platform. Garuda AI is its forecasting core. The separate Krishna Investigate lane handles unknown / unsupported evidence so the platform name and the triage function are not confused.

01 / END-TO-END ARCHITECTURE

PCAP / Flow → Feature Extraction → Graph State → GNN → LSTM → Future Trajectory → MITRE / Explainability → Controlled Response

01 / INGEST **PCAP + FLOW** Classic PCAP, flow CSV, NetFlow/IPFIX telemetry	**02 / EXTRACT** **FEATURE PIPELINE** Flow + packet features, 34 metrics, 10s graph windows	**03 / STATE** **GRAPH STATE** Host/service/protocol graph, node-edge summaries	**04 / SPATIAL** **GNN / GraphSAGE** Relational message passing, spatial embedding	**05 / TEMPORAL** **GARUDA AI · LSTM** Learns P(S_t+1 | S_t) from 8 observed windows (~80s)

06 / FORWARD SIMULATION **AUTOREGRESSIVE ROLLOUT** Predicted state feeds next step: +10s → +20s → +30s → +40s forward state + uncertainty + risk	**07 / INTERPRET** **STAGE + EXPLAINABILITY** MITRE-aligned stage only when supervised support exists; local gradient×input feature attribution	**08 / DECISION** **CONTROLLED RESPONSE ROUTER** Separates known response, unknown triage and signed containment. High risk alone never authorizes.

ARJUNA Known / reviewed policy-gated block	**KRISHNA INVESTIGATE** Unknown / unsupported triage + evidence	**SUDARSHANA** Authorized containment signed + expiry

WHY THIS IS A WORLD MODEL

- Models an ordered network state instead of classifying flows independently.
- Learns P(S_t+1 | S_t) and performs autoregressive K-step forward simulation.
- Keeps forecasting, stage interpretation and actuation as separate evidence lanes.
- Abstains / stays shadow-only when runtime support is unresolved.

REFERENCE TECH STACK

- **Model:** Python 3.12 + NumPy reverse-mode GraphSAGE/GNN + LSTM (no heavy PyTorch in runtime).
- **Training/Eval:** scikit-learn, pandas, pytest; multi-seed experiments.
- **Serving/UI:** Python standard HTTP + authenticated bridge; HTML5/CSS3 console.
- **Audit/Control:** SQLite audit chain + HMAC-signed policy files; optional Node.js proxy.

02 / ONE INFERENCE CYCLE

Observed history to evidence in five explicit stages

1 OBSERVE 8 ordered graph windows ~80 s context	**2 VALIDATE** schema · chronology mode · support	**3 ENCODE** GNN spatial + LSTM temporal state	**4 FORECAST** +10/+20/+30/+40 s state + risk evidence	**5 EXPLAIN / ROUTE** drivers · stage status controlled response

03 / SUBSYSTEM RESPONSIBILITIES

Clear interfaces, outputs and safety boundaries

SUBSYSTEM	RESPONSIBILITY	OUTPUT / BOUNDARY
Telemetry + Parser	Ingests PCAP / flow telemetry; normalizes timestamps and packet/flow evidence into versioned schema.	Unsupported formats or missing required evidence fail explicitly.
State / Graph Builder	Creates 10-second graph snapshots and ordered histories from service / host / protocol relationships.	Sequence gaps break histories; future windows never enter observed context.
Garuda Core	GraphSAGE/GNN encodes relational state; LSTM learns temporal transition dynamics from observed history.	State forecasting remains separate from risk, stage and response claims.
Forward Simulation	Autoregressively rolls predicted state to +10 / +20 / +30 / +40 seconds.	Future state + uncertainty + risk evidence; not a guaranteed compromise path.
Interpretability / Stage	Ranks local feature drivers; MITRE-aligned stage is shown only when supervised support exists.	Unresolved / insufficient evidence is preferred to an invented stage.
Defence + Audit	Arjuna = reviewed known-response; Krishna Investigate = unknown triage; Sudarshana = explicitly authorized containment.	Human/policy gates, expiry, revoke/kill-switch and audit prevent blind auto-blocking.

04 / DEPLOYMENT, FEASIBILITY & SCALE

Offline-first enterprise / CII path with explicit prototype limits

DEPLOYMENT TOPOLOGY

Sensor / TAP / SPAN → local collector → graph builder → Garuda inference → analyst console → optional enforcement proxy.

- Inference runs fully offline; no cloud API is required.
- Local authenticated APIs keep runtime credentials server-side.
- Model/input hashes and versioned artifacts support reproducibility.
- Enterprise/CII deployment remains human-gated until customer-domain validation is complete.

FEASIBILITY + SCALE PATH

- Reference capacity controls: up to 32 graph nodes, ~250k CSV rows and ~10k occupied windows; larger networks require sensor/subnet partitioning rather than silent truncation.
- Measured compatible CPU runtime: **~2.9 ms mean forecast-only**, 332–349 sequential calls/s in-process, ~45 MiB max RSS.
- Scale path: parallel sensor workers + subnet/service graph partitions + central evidence dashboard; benchmark again on target hardware before deployment.

05 / HACK FOR SOCIAL CAUSE REQUIREMENT MAPPING

Problem-statement requirements mapped directly to implemented architecture

THEME REQUIREMENT	IMPLEMENTATION	VISIBLE EVIDENCE
Structured network state	10-second service/host/protocol graph snapshots with flow + packet evidence.	State / Graph Builder
Learn P(S_t+1 | S_t)	GraphSAGE spatial encoder + LSTM temporal transition model.	Garuda Core
K-step future simulation	Autoregressive +10/+20/+30/+40 s future-state rollout.	Forecast trajectory
MITRE + explainability	Stage output only when supervised support exists; gradient×input drivers.	Stage / attribution UI
Offline demo + baselines	Local PCAP/CSV inference; LR, LSTM and persistence comparisons.	Dashboard + evidence

06 / MEASURED EVIDENCE

Rounded, dataset-named results - separate claim lanes

CICAPT-IIOT2024
~27.5% lower state MSE
vs persistence on Phase-2 forecasting, 3/3 seeds

UNSW-NB15
~98.7% recall · ~0.29% FPR
frozen same-input independent-source; F1 ~94.8%

CPU RUNTIME
~2.9 ms mean
forecast-only compatible in-process lab benchmark

EVIDENCE DISCIPLINE

- Logistic Regression, LSTM and persistence baselines separate static classification from temporal/state-model value.
- Cross-domain portability is not claimed universally; unresolved domains enter shadow/abstain mode.
- Attack-step lead time is not relabelled as verified lead to successful compromise.

EVALUATOR TAKEAWAY

Krishna Defence is not a static IDS dashboard. It is a network-state simulation pipeline that learns how traffic evolves, forecasts multiple future states, explains the drivers, and connects that evidence to layered, human-gated cyber-defence controls.